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3. Critical points of 3x3 system

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Computation of critical points

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In the previous video, we derived three nonlinear

equations to describe the interactions between a population of elephant, a population of rhino, and a population of antelope.

For example, the equation describing the evolution

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Our system modeling three species competing for the same resources

$$\dot{x} = x - x^2 - axy - bxz = x(1 - x - ay - bz) \quad (5.29)$$

$$\dot{y} = y - y^2 - bxy - ayz = y(1 - y - bx - az) \quad (5.30)$$

$$\dot{z} = z - z^2 - axz - byz = z(1 - z - ax - by) \quad (5.31)$$

has $2^3 = 8$ critical points. Seven of these critical points are

Equation(s) satisfied	Critical point(s)
$x = y = z = 0$	$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$
$y = z = 0, \quad x = 1$	$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$
$x = z = 0, \quad y = 1$	$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$
$x = y = 0, \quad z = 1$	$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$
$z = 0, \quad \begin{cases} 1 - x - ay = 0 \\ 1 - y - bx = 0 \end{cases}, \quad ab \neq 1$	$\frac{1}{1-ab} \begin{pmatrix} 1-a \\ 1-b \\ 0 \end{pmatrix}$
$y = 0, \quad \begin{cases} 1 - x - bz = 0 \\ 1 - z - ax = 0 \end{cases}, \quad ab \neq 1$	$\frac{1}{1-ab} \begin{pmatrix} 1-b \\ 0 \\ 1-a \end{pmatrix}$
$x = 0, \quad \begin{cases} 1 - y - az = 0 \\ 1 - z - by = 0 \end{cases}, \quad ab \neq 1$	$\frac{1}{1-ab} \begin{pmatrix} 0 \\ 1-a \\ 1-b \end{pmatrix}$

	$(1 - b)$
$x, y, z \neq 0$	(Find as an exercise.)

Getting started with symbolic variables in MATLAB

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You've probably seen expressions like these before.

And while each originates from different areas of math, science, and engineering, they all have one thing in common, symbolic variables.

A symbolic variable is a quantity that doesn't yet have an assigned value, and is instead represented by, you guessed it, a symbol

Working with symbolic expressions and functions in MATLAB

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Finding a maximum or minimum value of a quantity is a common task in science, engineering, economics, and many other areas.

And while there are numerical tools which allow us to estimate these values, we can often find the exact values by applying symbolic methods.

Let's see how to use the Symbolic Math

3. Critical points of 3x3 system

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